HOMEWORK 3

1. A transitive G-action on a nonempty set X is called *simply transitive* if the stabilizers $Stab(x)$ are trivial for all $x \in X$.
(a) Prove that G acts simply transitively on itself by left translations.
(b) Prove that if G acts simply transitively on X, then X is G-isomorphic to G with the G-action by left translations.

2. Let G act on a non-empty set X. Consider the map $f : G \times X \to X \times X$ defined by $f(g, x) = (gx, x)$. Prove that the action is simply transitive if and only if f is a bijection.

3. Let G and H be two isomorphic groups. Show that there are two simply transitive actions of the groups $Aut(G)$ and $Aut(H)$ on the set X of all group isomorphisms between G and H.

4. Let G be a p-group and let k be a divisor of $|G|$.
(a) Prove that G contains a *normal* subgroup of order k.
(b) Prove that every group of order p^2 (for a prime p) is abelian.

5. Find all Sylow subgroups in $\mathbb{Z}/60\mathbb{Z}$.

6. Let $H \subset G$ be a subgroup and let $P \subset H$ be a Sylow p-subgroup in H. Prove that there is a Sylow p-subgroup $Q \subset G$ such that $Q \cap H = P$.

7. (a) A subgroup H of G is called *characteristic*, if $f(H) = H$ for every automorphism f of G. Show that a characteristic subgroup H is normal in G.
(b) Prove that if K is a characteristic subgroup of H and H is a characteristic subgroup of G, then K is characteristic in G.

8. (a) For any two subgroups K and H of a group G denote by $[K, H]$ the subgroup in G generated by the commutators $[k, h] = khk^{-1}h^{-1}$ for all $k \in K$ and $h \in H$. Show that if K and H are normal in G, then so is $[K, H]$.
(b) Prove that $[G, H]$ is normal in G for every subgroup $H \subset G$.

9. Assume that a subset $S \subset G$ of a group G satisfies $gSg^{-1} \subset S$ for all $g \in G$. Prove that the subgroup generated by S is normal in G.

10. Let N be an abelian normal subgroup of a finite group G. Assume that the orders $|G/N|$ and $|Aut(N)|$ are relatively prime. Prove that N is contained in the center of G.